

LET Sample Paper Two - RF 2026 – Marking Guide**Q.1 – No (2 Marks)****Clause 4.13.3.2 (2 Marks)****Q.2 – “ENSURE THAT CAPACITORS ARE COMPLETELY DISCHARGED BEFORE WORKING ON EQUIPMENT” (2 Marks)****Clause 4.15.3.1 (2 Marks)****Q.3 – Gas or Vapour and Dust (2 Marks)****Clause 7.7.2.2 (a) & (b) (2 Marks)****Q.4 – (a) made automatically, before the live connections are made, when any****(b) broken automatically, not before the live connections are broken, when any (2 Marks)****Clause 5.5.4.4 (a) (b) (2 Marks)****Q.5 – IP2X (2 Marks)****Clause 2.7.7 (a) (2 Marks)****Q.6 – shall be double insulation or the equivalent of double insulation (2 Marks)****Clause 2.5.8 (2 Marks)****Q.7 – 200mm (2 Marks)****Clause 219 (3) (2 Marks)****Q.8 – 1 – Ratio of 30:2 (4 Marks)****Q.9**Table 3(3) Item 4 Table 13 Col 23 – $270A/3 = 90A$ **(1 Mark)** *(Item number is optional do not deduct marks)*

Part (i)

Table 25(2) Col 5 Derating for more than one circuit = 0.88 **(1 Mark)**Table 28(1) Col 2 Derating for Depth 0.6m = 0.99 **(1 Mark)** $90A/(0.88 \times 0.99) = 103.3A$ **(1 Mark)** $= 25mm^2 @ 125A$ **(1 Marks)**

Part (ii)

0.15m spacing Table 25(3) Col 3 Derate = 0.78 **(1 Mark)** $90A/(0.78 \times 0.99) = 116.55A$ **(1 Mark)** $= 25mm^2 @ 125A$ **(1 Marks)****(Deduct 1 Mark for no or incorrect units)****Q.10**

(i) 320mA

(ii) 96W

(iii) 62.4V

(Deduct 1 Mark for no or incorrect units)

Q.11**Table C1 Column 2 (1 Mark)**

Equipment	Load Group	Calculation	Maximum Demand
27 lighting points	(a)(i)	3A – 1 st 20 points plus 2A for the remaining 7 points 3+2=5A	5A (2 Marks)
1500W tennis court lighting	(a)(ii)	$1500/230 \times 75\% = 4.89A$	4.89A (1 Mark)
2.8kW spa pool heater	(g)	$2800/230 \times 75\% = 9.13A$	9.13A (1 Mark)
81 – 10A socket outlets (36 Doubles + 9 Singles)	(b)(i)	10A – 1 st 20 outlets + 5A for the next 20 + for 5A the next 20 + 5A for the next 20 + 5A for remaining 1 outlet $10 + 5 + 5 + 5 + 5 = 30A$	30A (1 Mark)
1 – 20A	(b)(iii)	15A	15A
9kW Range	(c)	$9000/230 \times 50\% = 19.57A$	19.57A
		Total Maximum Demand	83.59A (1 Marks)

(Deduct 1 Mark for no or incorrect units)**Q.12****Consumers Mains**

Table 41 Col. 10 (1 Mark)

 $V_c = 0.491$ (1 Mark) $120 \times 8 \times 0.491 / 1000 = 0.491$ (1 Mark) $V_d = 0.47V$ **Sub-mains**

Table 41 Col. 6 (1 Mark)

 $V_c = 1.55$ (1 Mark) $60 \times 110 \times 1.55 / 1000$ $V_d = 10.23V$ (1 Mark)**Final sub-circuit**

Table 42 Col. 8 (1 Mark)

 $V_c = 10.92$ (1 Mark) $28 \times 43 \times 10.2 / 1000$ $V_d = 12.28V$ (1 Mark)**Total Voltage Drop** $0.47 + 10.23 + 12.28 = 22.98V$ (1 Mark)**(Deduct 1 Mark for no or incorrect units. Deduct 1 mark for no or incorrect table number)****Q.13**

Ratio/rating = 5 (1 Mark) (deduct 1 Mark if unit is given)

Minimum Time = Accept 1.2-1.5 seconds (1 Mark)

Maximum Time = Accept 4-4.5 seconds (1 Mark)

(Deduct 1 mark if unit is given for ratio or unit for no or incorrect time unit)

Q.14

Transformer impedance = 0.01236Ω (3 Marks)

MSB Fault Current = 13060.76A (3 Marks)

DB Fault Current – 4388.47A (3 Marks)

(Deduct 1 Mark from each answer if candidate has rounded transformer impedance to 4 decimal places instead of 5)

(Deduct 1 Mark for no or incorrect units)

Q.15

35A (3 Marks)

Q.16

D (2 Marks)

Q.17

AS 4202 (2 Marks)

Clause number – 8.5.2 (2 Marks)

Q.18

(2 marks for correct defect one mark for the correct clause)

(Only accept the first 5 defects candidate has listed)

1. Main earthing conductor is not connected to the main earth bar as required – **5.3.4 (b)** or **5.5.1.1**
2. Neutral conductor not provided for the fire pump control circuit – **7.2.5.6.4 (a)**
3. Fire pump main switch is not labelled “IN THE EVENT OF FIRE, DO NOT SWITCH OFF” – **7.2.4.4 (b)**
4. Distribution board main switch not rated to the max. demand current of 110A – **2.2.4.3 (a)**
5. Distribution board protective earthing conductor should be minimum of 25mm^2 – **5.3.3.1.2 (a)**

Table 5.1

6. Neutral bar connections not sequenced to the correct circuit number – **2.10.5.4**
7. Consumers mains cannot maintain supply when exposed to fire – **7.2.2.1**
8. Location of main earth electrode not marked on the main switchboard – **5.3.6.4**
9. Main switch for fire pump “ON/OFF” not labelled as required – **2.3.2.2.1 (c)**
10. Main earthing conductor is uninsulated above the ground – **5.3.2.4**
11. Main switch for distribution board is not identifying which portion of the installation it controls – **2.3.3.5 (b)** or **2.10.5.2**
12. Consumers mains are too small for the MD of the installation – **3.4.1** or **1.6.3**
13. Consumers mains not protected for overcurrent at the main switchboard – **2.5.1.2 (b)**
14. Fire pump cables is undersize for the load – **3.4.1** or **Table 19 Col. 4 (AS/NZS 3008)**